

NEGRESTI, Romania

WMO: 151130

Lat: 46.8381N

Lon: 27.4422E

Elev: 131

StdP: 99.76

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.4	(c) -12.4	(d) -17.0	(e) 0.9	(f) -14.7	(g) -14.4	(h) 1.1	(i) -11.8	(j) 13.7	(k) -6.9	(l) 11.7	(m) -6.0	(n) 1.1	(o) 320	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.0	(c) 32.5	(d) 20.1	(e) 31.0	(f) 19.9	(g) 29.5	(h) 19.6	(i) 22.6	(j) 28.8	(k) 21.7	(l) 27.9	(m) 20.8	(n) 26.8	(o) 3.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.6	25.2	19.7	14.7	23.9	18.9	13.9	22.7	67.1	28.6	63.8	28.0	60.6	26.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.7	(b) 8.5	(c) 7.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 9.7	(b) 8.5	(c) 7.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-3.3	0.0	4.3	10.1	16.2	20.1	21.5	21.4	16.9	10.2	5.5	-0.2		
	DBStd	9.55	4.60	5.18	5.17	4.43	3.46	2.93	2.79	3.17	4.48	3.89	4.56	4.76		
	HDD10.0	1446	414	280	184	55	3	0	0	0	3	45	146	316		
	HDD18.3	3272	672	513	436	247	83	16	7	10	78	252	384	574		
	CDD10.0	1543	0	0	7	58	194	303	355	353	209	52	12	0		
	CDD18.3	328	0	0	0	1	16	69	103	104	34	1	0	0		
	CDH23.3	3483	0	0	1	20	219	689	988	1127	428	11	1	0		
	CDH26.7	1221	0	0	0	2	48	206	353	471	141	1	0	0		
(14) Wind	WSAvg	2.8	2.9	2.9	3.1	3.5	2.9	2.9	2.5	2.5	2.5	2.6	2.7	2.7		
(15) Precipitation	PrecAvg	527	22	21	28	43	59	82	71	46	48	47	32	29		
	PrecMax	775	49	54	63	94	181	173	176	110	155	197	62	85		
	PrecMin	367	5	2	1	4	6	17	7	14	7	7	1	2		
	PrecStd	110	14	13	15	21	40	38	42	27	40	39	20	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	15.2	21.7	25.6	30.5	32.2	34.0	34.7	32.2	25.0	20.1	12.1		
		MCWB	5.8	9.0	13.4	15.9	18.4	22.0	20.1	21.2	19.6	16.6	14.1	8.9		
	2%	DB	5.6	11.3	17.8	23.4	27.8	30.5	32.1	32.7	30.0	21.9	16.7	10.2		
		MCWB	3.8	6.9	11.1	14.4	17.8	21.2	19.7	19.8	18.5	14.7	12.3	8.2		
	5%	DB	4.1	9.1	14.8	21.1	25.9	29.0	30.3	31.2	28.2	19.8	14.0	8.4		
		MCWB	2.5	5.6	9.3	13.0	16.9	20.3	20.0	19.6	17.4	14.0	11.2	6.7		
	10%	DB	2.7	6.9	12.1	18.9	23.8	27.5	28.6	29.5	25.7	17.2	11.7	5.7		
		MCWB	1.5	4.7	7.7	12.4	16.0	19.6	19.4	19.4	16.9	12.8	9.9	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	9.6	13.7	16.7	20.1	24.2	23.1	23.7	20.6	17.8	14.7	10.5		
		MCDB	7.4	13.9	20.3	24.9	27.0	30.2	29.5	31.1	29.0	22.6	19.3	11.3		
	2%	WB	4.4	7.3	11.5	15.2	19.0	22.8	22.0	22.2	19.4	15.9	13.0	8.3		
		MCDB	5.5	10.4	16.9	22.0	26.2	28.5	28.4	29.1	26.9	20.6	16.0	9.6		
	5%	WB	2.8	6.1	9.9	13.9	17.8	21.8	21.2	21.4	18.6	14.6	11.6	6.6		
		MCDB	3.8	8.2	14.1	19.9	23.8	27.4	27.3	27.8	25.2	18.5	14.0	8.4		
	10%	WB	1.6	4.7	8.3	12.6	16.8	20.6	20.4	20.5	17.7	13.2	10.1	4.4		
		MCDB	2.4	6.7	11.7	17.8	22.0	26.0	26.2	26.5	24.0	16.7	11.8	5.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	8.0	9.9	11.4	12.1	11.8	12.0	13.0	12.7	10.6	7.1	6.1		
		MCDBR	8.1	11.6	15.5	16.2	15.4	14.4	15.1	16.4	16.8	16.0	12.1	9.1		
	5% WB	MCWBR	6.8	7.4	9.0	8.5	6.6	6.0	5.3	5.3	6.7	9.1	7.1	7.4		
		MCWBR	7.3	9.6	14.7	14.7	13.2	13.2	12.8	13.9	15.0	14.4	9.9	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.357	0.394	0.453	0.424	0.421	0.447	0.456	0.412	0.378	0.352	0.324		
	taud		2.266	2.225	2.194	2.062	2.222	2.266	2.201	2.167	2.248	2.321	2.332	2.308		
	Ebn at Noon		740	792	818	800	843	846	816	790	791	762	706	697		
	Edh at Noon		80	103	123	154	136	131	138	138	116	93	76	69		
(44) All-Sky Solar Radiation	RadAvg		1.25	2.01	3.20	4.39	5.62	5.90	5.95	5.40	3.82	2.45	1.29	0.98		
	RadStd		0.17	0.26	0.33	0.39	0.48	0.52	0.48	0.41	0.41	0.29	0.12	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		+0.64	N/A	N/A	+0.72	N/A	N/A	-134	-192	+98	+37

Nomenclature: See separate page